ORIGINAL ARTICLE

Updated Overall Survival Analysis From IMpower110: Atezolizumab Versus Platinum-Based Chemotherapy in Treatment-Naive Programmed Death-Ligand 1–Selected NSCLC

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ABSTRACT

Introduction: IMpower110 previously revealed significant overall survival (OS) benefit with atezolizumab versus chemotherapy in patients with treatment-naive EGFR- and ALK-negative (wild type [WT]) metastatic NSCLC with high programmed death-ligand 1 (PD-L1) expression (≥50% on tumor cells [TCs] or ≥10% on tumor-infiltrating immune cells [ICs], per SP142 immunohistochemistry assay; $p = 0.0106$). We present primary OS analyses in lower PD-L1 expression groups and an updated, exploratory analysis in the high PD-L1 expression group.

Methods: This open-label, phase 3 trial randomized patients with PD-L1 expression on greater than or equal to 1% of TC or IC to receive atezolizumab or platinum-based chemotherapy. The primary end point was OS, hierarchically tested in PD-L1 expression WT subgroups: first the high PD-L1 expression subgroup, then the high-or-intermediate PD-L1 expression subgroup (≥5% on TC or IC), and then the any PD-L1 expression subgroup (≥1% on TC or IC).

Results: The any PD-L1 expression WT population included 554 patients (excluded 18 EGFR- or ALK-positive patients). With 17 months' additional follow-up, OS improvement in the atezolizumab versus chemotherapy arm was not statistically significant in high-or-intermediate PD-L1 expression WT patients (n = 328; hazard ratio = 0.87, 95% confidence interval: 0.66–1.14, $p = 0.3091$; median = 19.9 versus 16.1 mo), precluding formal OS testing in any PD-L1 expression WT patients. Exploratory analysis in high PD-L1 expression WT patients (n = 205) revealed maintained OS benefit in the atezolizumab arm (hazard ratio = 0.76, 95% confidence interval: 0.54–1.09; median = 20.2 versus 14.7 mo). Updated safety data continued to favor atezolizumab.

Conclusions: Statistical significance for OS was not revealed in the high-or-intermediate expression WT group, and, as a result, OS in the any PD-L1 expression WT group was not formally tested. No new safety signals were found. This updated analysis of IMpower110 supports using atezolizumab in treatment-naive, metastatic WT NSCLC with high PD-L1 expression.

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Keywords: Atezolizumab; Chemotherapy; PD-L1; NSCLC; IMpower110

Introduction

The immune checkpoint molecules programmed cell death protein-1 and programmed death-ligand 1 (PD-L1) are important therapeutic targets in advanced NSCLC without targetable alterations in *EGFR* or *ALK* (wild type [WT]). On the basis of overall survival (OS) benefits, first-line standard-of-care therapy now includes checkpoint inhibitor monotherapy, checkpoint inhibitors combined with chemotherapy, with or without bevacizumab, or dual checkpoint inhibitor combination therapy.[1–9]

At the interim OS analysis of the phase 3 IMpower110 trial, atezolizumab (anti–PD-L1 antibody) was found to have statistically significant and clinically meaningful OS improvement versus platinum-based chemotherapy in patients with WT tumors and high PD-L1 expression (≥50% expression on tumor cells [TCs] or ≥10% expression on tumor-infiltrating immune cells [ICs], per SP142 immunohistochemistry [IHC] assay): hazard ratio (HR) = 0.59, 95% confidence interval (CI): 0.40–0.89; $p = 0.0106$; median = OS, 20.2 versus 13.1 months.[10] On the basis of these results, atezolizumab is approved in the United States as first-line treatment for WT NSCLC with high PD-L1 expression.

At that analysis, OS benefit was not statistically significant in patients with WT tumors and high-or-intermediate PD-L1 expression (≥5% expression on TC or IC) by SP142 assay ($p = 0.0416$); thus, on the basis of

the prespecified testing hierarchy, formal testing of OS was not performed in patients with WT tumors and any PD-L1 expression (≥1% on TC or IC).

We present the final OS analysis in the high-or-intermediate and any PD-L1 expression WT groups from IMpower110. As the OS benefit with atezolizumab in the high PD-L1 expression group crossed the boundary for significance at the interim OS analysis, that is considered the primary analysis for that population. Thus, the updated OS analysis for the high PD-L1 expression group presented here is exploratory and not formally tested. Other efficacy end points, biomarker analyses, and safety findings are also updated.

Materials and Methods

Study Design and Patients

IMpower110 is an open-label, randomized, phase 3 trial (NCT02409342), performed according to the Good Clinical Practice guidelines and Declaration of Helsinki. Protocol approval was provided by independent ethics committees at each study center; all patients provided written informed consent.

Detailed eligibility and methodology information has been published previously.[10] In brief, patients with chemotherapy-naive stage IV nonsquamous or squamous NSCLC and measurable disease per the Response Evaluation Criteria in Solid Tumors version 1.1 were enrolled. PD-L1 expression per SP142 IHC assay (Ventana) was required on greater than or equal to 1% of TC or on IC covering greater than or equal to 1% of the tumor area.[11]

Though originally eligible, patients with sensitizing *EGFR* mutations or *ALK* translocations and previous targeted therapy were later excluded.[10] Such patients who had already been enrolled were excluded from efficacy analyses.

Treatment

Patients were randomized 1:1 to receive atezolizumab monotherapy or platinum-based chemotherapy, with stratification by sex (male versus female), Eastern Cooperative Oncology Group performance status (0 versus 1), histology (nonsquamous versus squamous), and PD-L1 expression status (≥1% PD-L1 on TC and any level of PD-L1 on IC versus <1% PD-L1 on TC and ≥1% PD-L1 on IC).

Treatments were administered intravenously on day 1 of each 21-day cycle. In the atezolizumab arm, treatment (1200 mg) continued until loss of clinical benefit, unacceptable toxicity, or death. In the chemotherapy arm, patients with nonsquamous disease were treated with cisplatin 75 mg/m^2 or carboplatin area under the concentration−time curve (area under the curve) 6 plus pemetrexed 500 mg/m^2 for four or six cycles and then pemetrexed maintenance therapy; those with squamous NSCLC received cisplatin 75 mg/m^2 plus gemcitabine 1250 mg/m^2 or carboplatin area under the curve 5 plus gemcitabine 1000 mg/m^2 for four or six cycles and then best supportive care.

Crossover to atezolizumab from the chemotherapy arm was not permitted. After study treatment discontinuation, patients could receive subsequent anticancer therapy outside the study protocol.

End Points and Assessments

OS was the primary study end point in three PD-L1–selected WT populations (tested hierarchically in the high [≥50% expression on TC or ≥10% expression on IC], then high-or-intermediate [≥5% expression on TC or IC], and then any PD-L1 [≥1% on TC or IC] expression groups) identified by the SP142 assay. Key secondary efficacy end points were investigator-assessed progression-free survival (PFS; per Response Evaluation Criteria in Solid Tumors version 1.1), objective response rate (ORR), duration of response (DOR), OS, and investigator-assessed PFS in PD-L1 expression subgroups per SP263 IHC assay and blood tumor mutational burden (bTMB) subgroups. Exploratory end points included OS and investigator-assessed PFS in prespecified PD-L1 expression subgroups per 22C3 IHC assay.

Tumor assessments were conducted at baseline and every 6 weeks for 48 weeks and then every 9 weeks until the first occurrence of radiographic disease progression (or loss of clinical benefit for atezolizumab-arm patients treated beyond progression), consent withdrawal, or death, whichever occurred first.

The SP263 (Ventana, Tuscon, AZ),[12] 22C3 (Dako, Carpinteria, CA),[13] and bTMB (Foundation Medicine, Cambridge, MA)[14] assays were performed as described previously.[10]

The safety population comprised all patients who received the study treatment, irrespective of PD-L1 expression and EGFR or ALK mutational status. Adverse events (AEs) were graded using the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.0 and coded according to Medical Dictionary for Regulatory Activities version 22.1. AEs were reported until 30 days (90 d for serious AEs and AEs of special interest) after the last dose of the study drug or the initiation of new systemic anticancer therapy, whichever occurred sooner.

Statistical Analysis

To control for the overall type I error rate at a two-sided significance level of 0.05, the primary OS end point was tested hierarchically in the WT population:

first, the high PD-L1 expression subgroup, then the high-or-intermediate PD-L1 expression subgroup, and then the any PD-L1 expression subgroup. If OS was statistically positive in all three populations, a two-sided significance level of 0.05 was passed down for analysis of PFS.

Because the prespecified alpha boundary was crossed in the high PD-L1 expression WT group at the interim OS analysis,[10] that became the primary analysis of OS in that group; thus, further analyses were exploratory only and not formally tested for the high PD-L1 WT population. The final OS analysis for the remaining groups was performed after approximately 216 and 390 deaths had occurred in the high-or-intermediate PD-L1 and any PD-L1 expression WT groups, respectively. At that time, an updated, exploratory analysis in the high PD-L1 expression WT group was also conducted.

A stratified log-rank test was used to analyze OS and PFS, with HRs and 95% CIs estimated using a stratified Cox regression model. Medians and their 95% CIs were estimated using the Kaplan-Meier and Brookmeyer-Crowley methodologies, respectively. Prespecified subgroup analyses were carried out to determine the consistency of the treatment effect using unstratified HRs estimated from a Cox proportional hazards model. ORR and its 95% CI were calculated using the Clopper-Pearson method. DOR was estimated using Kaplan-Meier methodology.

Table 1. Baseline Demographic and Patient Characteristics

	Any PD-L1 Expression WT		High-or-Intermediate PD-L1 Expression WT		High PD-L1 Expression WT	
Characteristics	Atezo, n = 277	Chemo, n = 277	Atezo, n = 166	Chemo, n = 162	Atezo, n = 107	Chemo, n = 98
Age, y						
Median	64	65	63	65	63	66
Range	30-81	30-87	33-81	33-87	33-79	33-87
Age group, y, n (%)						
<65	143 (51.6)	134 (48.4)	93 (56.0)	76 (46.9)	59 (55.1)	43 (43.9)
65-74	106 (38.3)	117 (42.2)	50 (30.1)	75 (46.3)	33 (30.8)	47 (48.0)
>74	28 (10.1)	26 (9.4)	23 (13.9)	11 (6.8)	15 (14.0)	8 (8.2)
Sex, n (%)						
Male	196 (70.8)	193 (69.7)	122 (73.5)	107 (66.0)	79 (73.8)	64 (65.3)
Female	81 (29.2)	84 (30.3)	44 (26.5)	55 (34.0)	28 (26.2)	34 (34.7)
Race, n (%)						
White	227 (81.9)	240 (86.6)	133 (80.1)	139 (85.8)	87 (81.3)	82 (83.7)
Asian	45 (16.2)	30 (10.8)	31 (18.7)	20 (12.3)	20 (18.7)	15 (15.3)
Black or African American	2 (0.7)	2 (0.7)	1 (0.6)	0	0	0
Multiple	1 (0.4)	0	0	0	0	0
Unknown	2 (0.7)	5 (1.8)	1 (0.6)	3 (1.9)	0	1 (1.0)
Region, n (%)						
Europe	209 (75.5)	225 (81.2)	118 (71.1)	131 (80.9)	76 (71.0)	77 (78.6)
Asia Pacific	44 (15.9)	29 (10.5)	31 (18.7)	19 (11.7)	20 (18.7)	14 (14.3)
South America	15 (5.4)	16 (5.8)	11 (6.6)	7 (4.3)	6 (5.6)	5 (5.1)
North America	9 (3.2)	7 (2.5)	6 (3.6)	5 (3.1)	5 (4.7)	2 (2.0)
ECOG performance status, n (%)[a]						
0	97 (35.0)	103 (37.2)	60 (36.1)	63 (38.9)	35 (32.7)	39 (39.8)
1	180 (65.0)	174 (62.8)	106 (63.9)	99 (61.1)	72 (67.3)	59 (60.2)
Tobacco use history, n (%)						
Never	37 (13.4)	35 (12.6)	21 (12.7)	17 (10.5)	9 (8.4)	15 (15.3)
Current	74 (26.7)	81 (29.2)	38 (22.9)	52 (32.1)	20 (18.7)	29 (29.6)
Previous	166 (59.9)	161 (58.1)	107 (64.5)	93 (57.4)	78 (72.9)	54 (55.1)
Histology at diagnosis, n (%)						
Nonsquamous	192 (69.3)	193 (69.7)	122 (73.5)	116 (71.6)	80 (74.8)	75 (76.5)
Squamous	85 (30.7)	84 (30.3)	44 (26.5)	46 (28.4)	27 (25.2)	23 (23.5)

Note: High: PD-L1 expression greater than or equal to 50% TC or greater than or equal to 10% IC; high-or-intermediate: PD-L1 expression greater than or equal to 5% TC or IC; any: PD-L1 expression greater than or equal to 1% TC or IC.

Atezo, atezolizumab; Chemo, chemotherapy; ECOG, Eastern Cooperative Oncology Group; IC, tumor-infiltrating immune cell; PD-L1, programmed death-ligand 1; TC, tumor cell; WT, wild type.

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[a]Subsequent to the interim analysis (data cutoff: September 10, 2018), one patient in the chemotherapy arm was reclassified from an ECOG performance status of 1 to 0.

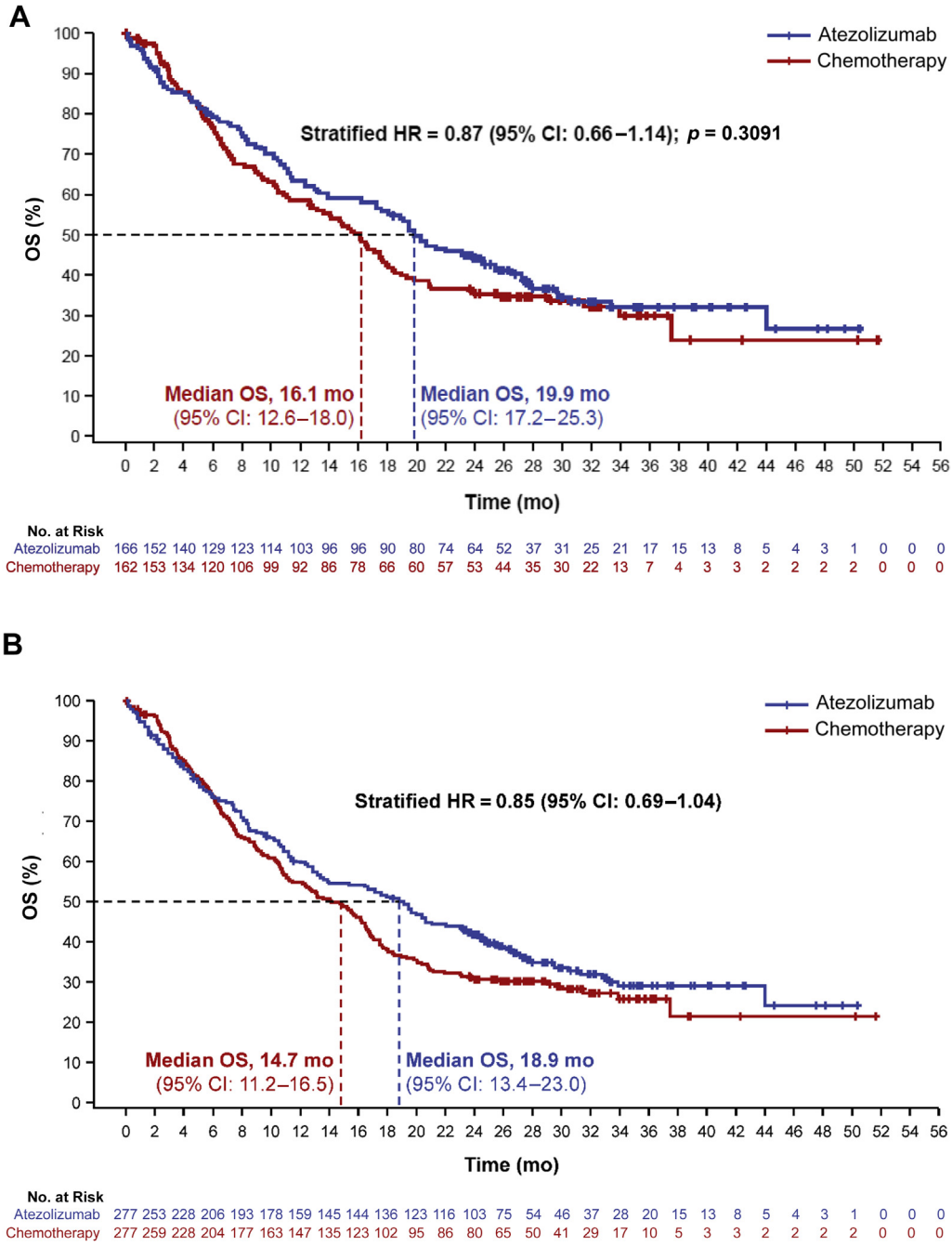

Figure 1. Kaplan-Meier estimates of OS in (*A*) high-or-intermediate PD-L1 expression (≥5% PD-L1 expression on TC or IC) WT group and (*B*) any PD-L1 expression (≥1% PD-L1 expression on TC or IC) WT group. CI, confidence interval; HR, hazard ratio; IC, tumor-infiltrating immune cell; OS, overall survival; PD-L1, programmed death-ligand 1; TC, tumor cell; WT, wild type.

An additional post hoc exploratory sensitivity analysis for OS was performed using a rank-preserving structural failure time model to adjust for the effect of subsequent immunotherapy as nonprotocol therapy (NPT).[15] OS duration after the start of NPT immunotherapy for patients in the chemotherapy arm was adjusted multiplicatively by a given factor to estimate the OS in the absence of NPT. The total OS duration—the sum of the time from randomization to the start of NPT immunotherapy and the adjusted survival time after the start of NPT—was analyzed using the same methodology as the primary OS analysis.

Results

Patients

As reported previously, 572 patients were randomized (July 2015–February 2018), 285 to atezolizumab

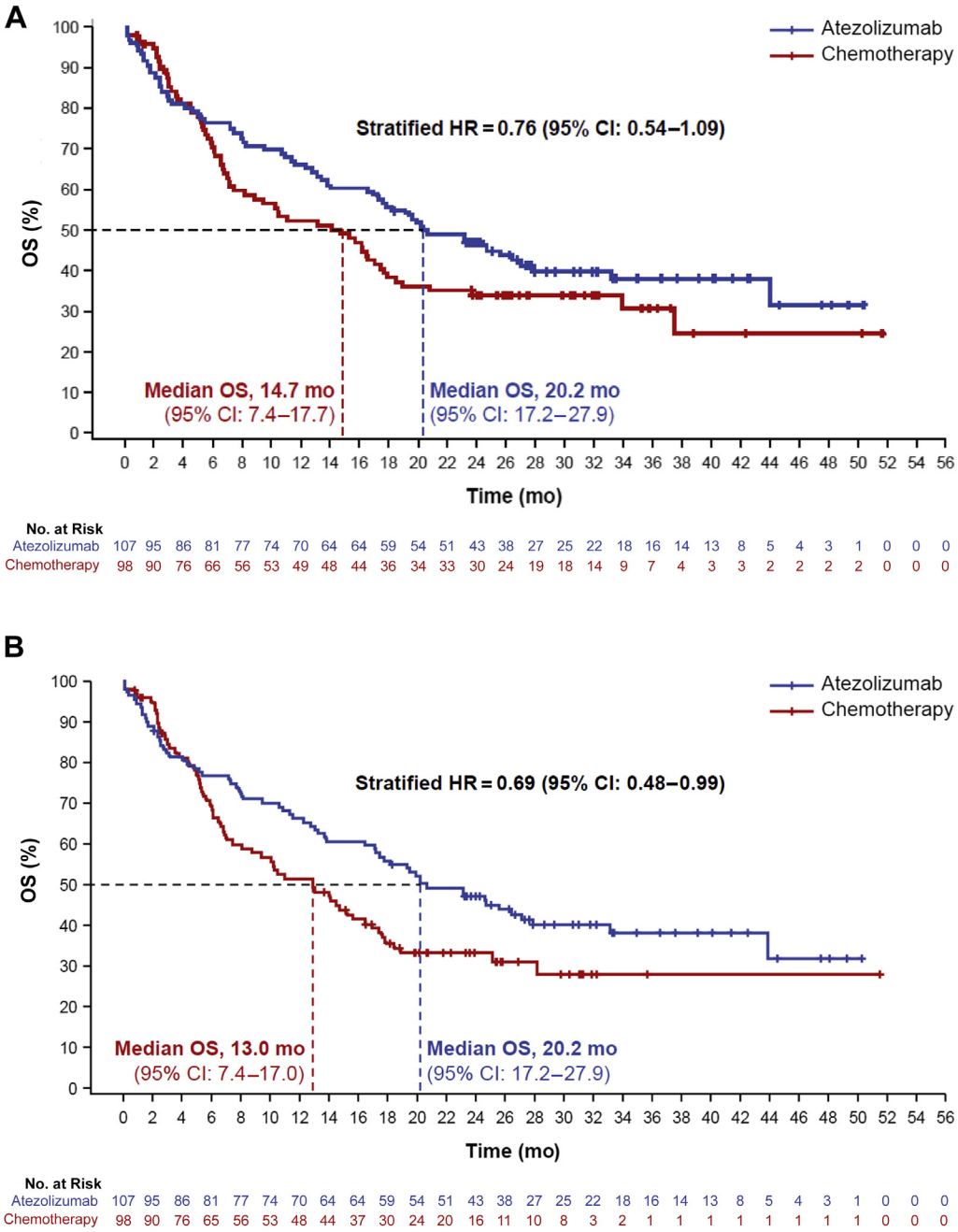

Figure 2. Kaplan-Meier estimates of OS in (*A*) the high PD-L1 expression (≥50% PD-L1 expression on TC or ≥10% PD-L1 expression on IC) WT group (exploratory analysis) and (*B*) the high PD-L1 expression WT group, adjusting for NPT immunotherapy using the RPSFT method. CI, confidence interval; HR, hazard ratio; IC, tumor-infiltrating immune cell; NPT, nonprotocol therapy; OS, overall survival; PD-L1, programmed death-ligand 1; RPSFT, rank-preserving structural failure time; TC, tumor cell; WT, wild type.

and 287 to chemotherapy. A total of 18 patients with *EGFR* or *ALK* alterations in their tumors were excluded; thus, the primary analysis population (any PD-L1 expression WT group) included 554 patients (277 per arm; Supplementary Fig. A1).[10] Overall, baseline characteristics were balanced between arms (Table 1).

The high PD-L1 expression WT group included 107 (38.6%) and 98 patients (35.4%) in the atezolizumab and chemotherapy arms, respectively; the high-or-intermediate PD-L1 expression WT group comprised 166 (59.9%) and 162 patients (58.5%) in the respective treatment arms.

Overall Survival

At the cutoff date (February 4, 2020), median OS follow-up was 31.3 (range: 0–52), 31.0 (range: 0–52), and 30.0 months (range: 0–52) in the high, high-or-

Table 2. Investigator-Assessed ORR, DOR, and PFS

Outcome	Atezo	Chemo
High PD-L1 expression WT	n = 107	n = 98
Median PFS, mo	8.2	5.0
HR (95% CI)	0.59 (0.43-0.81)	
1-y PFS, %	39.2	19.2
2-y PFS, %	25.7	7.9
ORR (95% CI), %	40.2 (30.8-50.1)	28.6 (19.9-38.6)
Median DOR (range), mo	38.9 (2.8-46.3[a])	8.3 (2.6-30.0[a])
High-or-intermediate PD-L1 expression WT	n = 166	n = 162
Median PFS, mo	7.3	5.5
HR (95% CI)	0.64 (0.50-0.82)	
1-y PFS, %	37.0	17.7
2-y PFS, %	22.0	6.3
ORR (95% CI), %	33.7 (26.6-41.5)	32.1 (25.0-39.9)
Median DOR (range), mo	38.9 (2.8-46.3[a])	5.8 (2.6-31.2[a])
Any PD-L1 expression WT	n = 277	n = 277
Median PFS, mo	5.8	5.6
HR (95% CI)	0.72 (0.60-0.86)	
1-y PFS, %	32.8	17.5
2-y PFS, %	18.6	5.2
ORR (95% CI), %	31.4 (26.0-37.2)	32.1 (26.7-38.0)
Median DOR (range), mo	26.3 (2.1-46.3[a])	5.7 (2.4-31.2[a])

Note: High: PD-L1 expression greater than or equal to 50% TC or greater than or equal to 10% IC; high-or-intermediate: PD-L1 expression greater than or equal to 5% TC or IC; any: PD-L1 expression greater than or equal to 1% TC or IC.
Atezo, atezolizumab; Chemo, chemotherapy; CI, confidence interval; DOR, duration of response; HR, hazard ratio; IC, tumor-infiltrating immune cell; ORR, objective response rate; PD-L1, programmed death-ligand 1; PFS, progression-free survival; TC, tumor cell; WT, wild type.
[a]Censored.

intermediate, and any PD-L1 expression WT groups, respectively. Deaths had occurred in 128 (61.8%), 211 (64.3%), and 370 patients (66.8%), respectively.

After 17 months of additional follow-up since the interim analysis, OS benefit in the atezolizumab versus chemotherapy arm in the high-or-intermediate PD-L1 expression WT group was not statistically significant (stratified HR = 0.87, 95% CI: 0.66–1.14, $p = 0.3091$; median = 19.9 versus 16.1 mo) (Fig. 1*A*). As a result, and per the statistical plan, OS was not formally tested in the any PD-L1 expression WT group. The median OS in that group was 18.9 months with atezolizumab and 14.7 months with chemotherapy (stratified HR = 0.85, 95% CI: 0.69–1.04) (Fig. 1*B*).

In the exploratory updated OS analysis in the high PD-L1 expression WT group, continued OS improvement in the atezolizumab versus chemotherapy arm was found (stratified HR = 0.76, 95% CI: 0.54–1.09; median = 20.2 versus 14.7 mo) (Fig. 2*A*). OS outcomes continued to favor atezolizumab across most key subgroups (Supplementary Fig. A2).

Similar proportions of patients across PD-L1 expression groups in the WT population received different classes of subsequent NPT (Supplementary Table A1). In the any PD-L1 expression WT group, fewer patients received any NPT in the atezolizumab arm (40.4%) than in the chemotherapy arm (58.8%), and 4.7% and 34.3% of these patients received NPT immunotherapy, respectively. In the high PD-L1 expression WT group, 3.7% in the atezolizumab arm and 34.7% in the chemotherapy arm received NPT immunotherapy; median time to first NPT immunotherapy was 14.4 months (range: 12.7–20.2) and 6.0 months (range: 1.0–24.3), respectively, and median duration of first NPT immunotherapy was 1.9 months (range: 0–7.4 [ongoing]) and 24.4 months (range: 0–43.5 [ongoing]).

After adjusting for the effect of NPT immunotherapy, median OS in the high PD-L1 expression WT group was 20.2 months in the atezolizumab arm and 13.0 months in the chemotherapy arm (HR = 0.69, 95% CI: 0.48–0.99) (Fig. 2*B*).

Additional Efficacy End Points

In each PD-L1 expression group in the WT population, the median PFS was longer with atezolizumab than chemotherapy (Table 2 and Supplementary Fig. A3). The benefit from atezolizumab was greatest in the high PD-L1 expression group: a median PFS of 8.2 versus 5.0 months (stratified HR = 0.59, 95% CI: 0.43–0.81) and 1-year PFS estimates of 39.2% and 19.2% in the atezolizumab and chemotherapy arms, respectively.

Confirmed ORR was higher with atezolizumab than chemotherapy in the high PD-L1 expression WT group

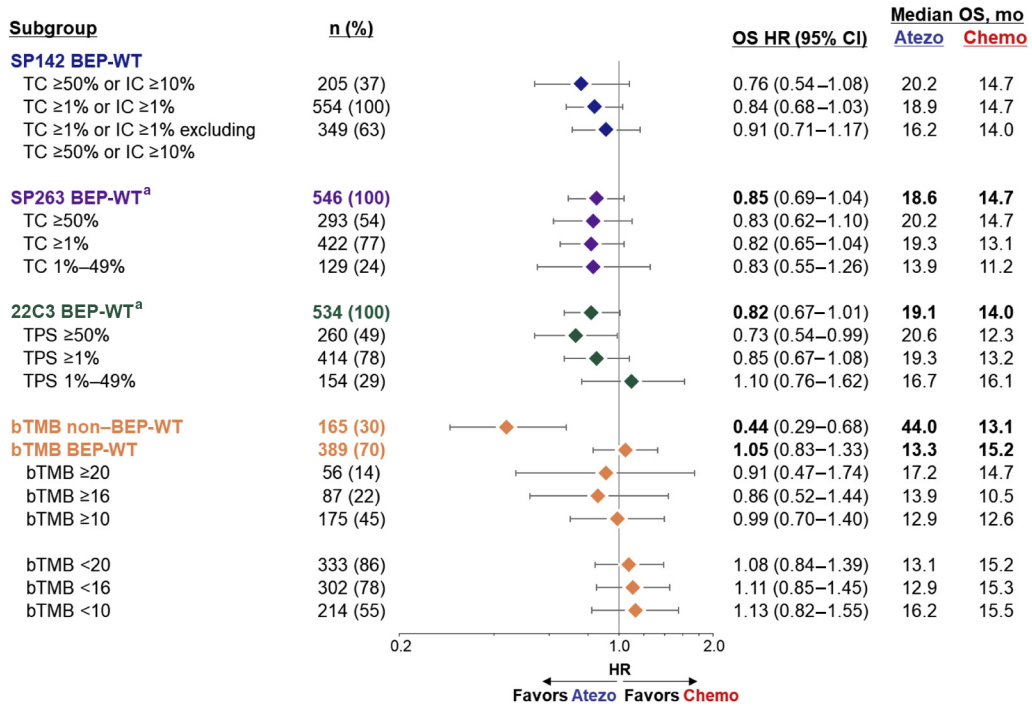

Figure 3. Forest plot of OS in SP263, 22C3, and bTMB subgroups within the SP263 BEP-WT, 22C3 BEP-WT, and bTMB BEP-WT populations, respectively. Unstratified hazard ratios are presented for all subgroups. [a]BEPs for the SP263 and 22C3 assays were similar to the any PD-L1 expression SP142 WT population; non-BEPs (not presented) included eight and 20 WT patients, respectively. Atezo, atezolizumab; BEP, biomarker-evaluable population; bTMB, blood tumor mutational burden; Chemo, chemotherapy; CI, confidence interval; HR, hazard ratio; IC, tumor-infiltrating immune cell; OS, overall survival; PD-L1, programmed death-ligand 1; TC, tumor cell; TPS, tumor proportion score; WT, wild type.

and similar in both arms in the other groups (Table 2). The median DOR, estimated in patients with a confirmed objective response, was longer with atezolizumab than chemotherapy in all three groups (Table 2).

Biomarker Analysis of Efficacy

The biomarker-evaluable WT populations for the SP263 and 22C3 assays comprised 546 and 534 patients, respectively; baseline patient characteristics in these

Table 3. Most Common AEs

	Atezo, n = 286			Chemo, n = 263		
AEs	All Grades	Grade 3/4	Grade 5	All Grades	Grade 3/4	Grade 5
Any all-cause AE	263 (92.0)	97 (33.9)	12 (4.2)	250 (95.1)	140 (53.2)	11 (4.2)
Anemia	49 (17.1)	5 (1.7)	0	127 (48.3)	50 (19.0)	0
Decreased appetite	49 (17.1)	2 (0.7)	0	51 (19.4)	0	0
Fatigue	44 (15.4)	3 (1.0)	0	47 (17.9)	6 (2.3)	0
Pyrexia	43 (15.0)	0	0	25 (9.5)	1 (0.4)	0
Nausea	42 (14.7)	1 (0.3)	0	89 (33.8)	5 (1.9)	0
Constipation	41 (14.3)	3 (1.0)	0	59 (22.4)	2 (0.8)	0
Asthenia	40 (14.0)	2 (0.7)	0	47 (17.9)	5 (1.9)	0
Pneumonia	22 (7.7)	10 (3.5)	0	20 (7.6)	11 (4.2)	1 (0.4)
Hyponatremia	19 (6.6)	8 (2.8)	0	12 (4.6)	6 (2.3)	0
Thrombocytopenia	8 (2.8)	1 (0.3)	0	45 (17.1)	20 (7.6)	0
Neutropenia	5 (1.7)	2 (0.7)	0	74 (28.1)	46 (17.5)	0
Hyperkalemia	13 (4.5)	7 (2.4)	0	10 (3.8)	4 (1.5)	0
Death	2 (0.7)	0	2 (0.7)	3 (1.1)	0	3 (1.1)
Decreased platelet count	1 (0.3)	0	0	22 (8.4)	11 (4.2)	0
Decreased neutrophil count	0	0	0	19 (7.2)	10 (3.8)	0
Febrile neutropenia	0	0	0	9 (3.4)	9 (3.4)	0

Note: Adverse events with an incidence of greater than or equal to 15% in any arm, grade 3/4 severity with incidence of greater than or equal to 2% in any arm, or grade 5 severity with an incidence of greater than or equal to 1% in any arm. Values given indicate number of patients (%).
AE, adverse event; Atezo, atezolizumab; Chemo, chemotherapy.

groups reflected those in the any PD-L1 expression WT population.[10] OS was generally comparable across equivalent PD-L1 expression WT subgroups defined by the three IHC assays for high PD-L1 (SP263 TC ≥50%; 22C3 TPS ≥50%; SP142 TC ≥50% or IC ≥10%), any PD-L1 (≥1% cutoffs for all three assays), and low PD-L1 expression (SP263 TC 1%–49%, 22C3 TPS 1%–49%, SP142 TC or IC ≥1% excluding TC ≥50% or IC ≥10%) (Fig. 3 and Supplementary Fig. A4). PFS benefit with atezolizumab versus chemotherapy also remained consistent across PD-L1 expression groups for the different assays (Supplementary Fig. A5).

The bTMB biomarker-evaluable WT populations included 389 patients and had similar baseline characteristics to the any PD-L1 expression WT population.[10] Similar to the findings of the interim analysis, OS benefit plateaued in the bTMB greater than or equal to 16 group, with a median OS of 13.9 and 10.5 months in the atezolizumab and chemotherapy arms, respectively (unstratified HR = 0.86, 95% CI: 0.52–1.44) (Fig. 3). The PFS in bTMB subgroups is presented in Supplementary Figure A5.

Safety

The safety population comprised 286 patients who received any amount of atezolizumab and 263 patients who received only chemotherapy. The median atezolizumab treatment duration was 5.3 months; median durations for cisplatin, carboplatin, gemcitabine, and pemetrexed were 2.1, 2.3, 2.6, and 3.5 months, respectively (Supplementary Table A2).

AEs were reported in 92.0% and 95.1% of atezolizumab- and chemotherapy-treated patients, respectively (Table 3). The most common AEs are reported in Table 3; AEs with a greater than 5% difference in incidence between treatment arms are illustrated in Supplementary Figure A6. Treatment-related AEs were found in 62.9% and 85.2% of atezolizumab- and chemotherapy-treated patients, respectively (Supplementary Table A2).

Grade 3/4 AEs occurred in 33.9% and 53.2% of patients in the atezolizumab and chemotherapy arms, respectively (Table 3); the most common in the safety population as a whole were anemia, neutropenia, and thrombocytopenia (all higher with chemotherapy). Serious AEs occurred in 31.8% and 29.3% of patients, respectively (Supplementary Table A2). One additional grade 5 AE was reported since the previous analysis (one patient with pulmonary edema died in the atezolizumab arm; the event was not immune mediated); a total of 12 (4.2%) and 11 patients (4.2%) in the atezolizumab and chemotherapy arms, respectively, died as a result of an AE (Table 3).

The rate of immune-mediated AEs was 46.2% with atezolizumab and 18.3% with chemotherapy; immune-mediated AEs were grade 3/4 in 8.7% and 1.5% of patients, respectively, and no grade 5 events of this type were found (Supplementary Table A2). Consistent with the previous analysis, the most frequent immune-mediated AEs in both arms were hepatic laboratory abnormalities and rash (Supplementary Table A3). Immune-mediated AEs requiring corticosteroid treatment are reported in Supplementary Table A4.

Discussion

At the updated OS analysis of IMpower110, a statistically significant OS improvement in the atezolizumab versus chemotherapy arm was not observed in the high-or-intermediate PD-L1 expression WT population ($p = 0.3091$), and therefore, the any PD-L1 expression WT population was not formally tested. According to the statistical analysis plan, the current analysis is the primary analysis for OS in both the high-or-intermediate and any PD-L1 expression WT populations.

In contrast, a significant OS improvement with atezolizumab in the high PD-L1 expression WT population was previously detected ($p = 0.0106$),[10] and that was therefore considered the primary OS analysis in that population. At this updated data cut, no formal statistical testing for OS in the high PD-L1 expression WT subgroup was performed. The exploratory updated analysis presented here supports the primary OS result and, with 17 months of additional follow-up, indicates that the OS benefit of atezolizumab is sustained in patients with tumors exhibiting high PD-L1 expression. OS outcomes favored atezolizumab in most key subgroups, although results from subgroups with small sample size should be interpreted with caution. Overall, these data provide additional support for the use of atezolizumab monotherapy in treatment-naive WT NSCLC with high levels of PD-L1.

At this updated data cut, median OS in the atezolizumab arm was similar to that from the primary analysis for the high PD-L1 expression subgroup; however, median OS increased in the chemotherapy arm. At the time of this analysis, in the high PD-L1 expression WT population, NPT immunotherapy was administered in 3.7% and 34.7% of patients in the atezolizumab and chemotherapy arms, respectively. Compared with the previous analysis, 5% more patients had received NPT immunotherapy in the chemotherapy arm. In addition to the higher frequency of NPT immunotherapy, chemotherapy-arm patients tended to start immunotherapy earlier and remain on it for longer than atezolizumab-arm patients (median time to first NPT immunotherapy: 14.4 mo in the atezolizumab arm versus 6.0 mo in the chemotherapy arm; median

duration of first NPT immunotherapy: 1.9 mo in the atezolizumab arm versus 24.4 mo in the chemotherapy arm), potentially affecting the measured survival benefit with atezolizumab with longer follow-up.

Checkpoint inhibitors have revealed survival benefit in NSCLC after progression on chemotherapy[16–18]; therefore, their use in this setting can dilute the treatment effect and complicate the interpretation of OS.[19,20] In contrast to OS, the secondary efficacy end points PFS, ORR, and DOR are not sensitive to the diluting effects of follow-up therapies. Compared with the previous analysis, in this analysis, PFS, ORR, and DOR differences between treatment arms in the high PD-L1 expression WT population remained substantial, suggesting the reduced OS benefit may have resulted from increased postprogression survival caused by higher NPT immunotherapy use in the chemotherapy arm. Indeed, the rank-preserving structural failure time sensitivity analysis used to estimate OS in the absence of NPT immunotherapy produced HR that more strongly favored atezolizumab. This indicates that at this updated analysis the use of NPT immunotherapy likely attenuated the observed OS effect, particularly in the high PD-L1 expression WT population. As mentioned previously, the use of subsequent anticancer agents was not regulated by the study protocol. Because of the observational nature of these data, they may be subject to confounding and bias and do not allow for firm conclusions. Despite this caveat, support comes from similar analyses of the impact of subsequent immunotherapy performed in the KEYNOTE-024 trial.[21] There, chemotherapy-arm patients with treatment-naive mutated NSCLC were permitted to cross over to pembrolizumab after disease progression, with a total subsequent immunotherapy rate of 64%. Additional analyses of OS adjusted for the impact of subsequent immunotherapy revealed that the estimated median OS in the chemotherapy arm would have been shorter in the absence of crossover.

The KEYNOTE-042 study revealed OS benefit for single-agent checkpoint inhibitor therapy in populations that included tumors with lower levels of PD-L1 expression.[5] Nevertheless, the OS HR in the low PD-L1 expression subgroup (22C3 TPS 1%–49%) in the final analysis of KEYNOTE-042 (0.91, 95% CI: 0.77–1.09)[22] closely resembled the outcomes found in low-expression subgroups detected by each of the three IHC assays in IMpower110, with the benefit in the overall population being driven by patients with high PD-L1 expression. Cross-trial comparisons of updated OS results should be made with caution owing to differences in follow-up duration between studies (median OS = 14 mo in KEYNOTE-042 versus >30 mo in IMpower110), among other factors.

Updated exploratory biomarker analyses using the SP263 and 22C3 assays revealed similar outcomes in PD-L1–high and –low subgroups compared with SP142-defined subgroups. Greatest PFS benefit with atezolizumab was observed in the bTMB greater than or equal to 16 and greater than or equal to 20 subgroups, similar to the interim analysis[10]; at the updated analysis, limited OS benefit was found only in the bTMB greater than or equal to 16 subgroup.

Safety in this updated analysis was consistent with that in the interim analysis and previous experience of atezolizumab monotherapy across indications, with no new or unexpected safety signals identified. Atezolizumab continued to reveal a more favorable safety profile than chemotherapy, including a lower frequency of grade 3/4 AEs and AEs leading to treatment withdrawal.

In conclusion, the updated OS analysis from IMpower110 confirms the OS benefit previously found with first-line atezolizumab monotherapy in patients with NSCLC whose tumors express high levels of PD-L1 (≥50% PD-L1 expression on TC or ≥10% on IC), supporting its use in this population. Statistically significant OS improvement was not observed in the high-or-intermediate PD-L1 expression population at this updated analysis, and as a result, OS in the any PD-L1 expression population could not be formally tested.

CRediT Authorship Contribution Statement

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Data Sharing

Qualified researchers may request access to individual patient-level data through the clinical study data request platform (https://vivli.org/). Further details on Roche's criteria for eligible studies are available here (https://vivli.org/members/ourmembers/). For further details on Roche's Global Policy on the Sharing of Clinical Information and how to request access to related clinical study documents, see here (https://www.roche.com/research_and_development/who_we_are_how_we_work/clinical_trials/our_commitment_to_data_sharing.htm).

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Supplementary Data

Note: To access the supplementary material accompanying this article, visit the online version of the *Journal of Thoracic Oncology* at www.jto.org/ and at https://doi.org/10.1016/j.jtho.2021.06.019.

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